MAMA'S PANTRY

Web Application

Technical Specification & Build Guide

Document Type	Technical Specification
Audience	Lead Developer / Engineering Team
Stack	Next.js 15 + Node.js + PostgreSQL
Version	1.0 — Initial Build
Date	2025 / 2026

1. Project Overview

Mama's Pantry requires two separate web applications built on a shared backend API. The first is a public-facing customer website (mamaspantry.com.ng) where Magodo residents and delivery customers browse products, place orders, and manage their accounts. The second is a private admin dashboard (admin.mamaspantry.com.ng) where the store owner and staff manage inventory, process orders, track deliveries, and view sales analytics.

Both applications share the same PostgreSQL database and Node.js/Express REST API. They are deployed separately on Vercel (frontend) and Railway or Render (backend).

Two Apps, One API

Customer Site: mamaspantry.com.ng — public, Next.js, SSR for SEO
Admin Dashboard: admin.mamaspantry.com.ng — private, React SPA, role-based access
Shared API: api.mamaspantry.com.ng — Node.js/Express, JWT auth, PostgreSQL

2. Technology Stack

2.1 Customer Website

Layer	Choice & Rationale
Framework	Next.js 15 (App Router). Server-side rendering for SEO, React Server Components for performance.
Language	TypeScript throughout. Strict mode enabled.
Styling	Tailwind CSS v4. Custom design tokens matching Mama's Pantry brand colours.
State Management	Zustand for client cart state. React Query (TanStack) for server state and caching.
Payment	Paystack inline SDK. Handles card, bank transfer, and USSD.
Auth	NextAuth.js v5 with email/password and Google OAuth. JWT sessions.
Image Optimisation	Next.js Image component with Cloudinary for product image hosting and transformation.
Fonts	Google Fonts: Playfair Display (display), Lora (body), DM Sans (UI). next/font for zero layout shift.
Hosting	Vercel. Preview deployments on every pull request.
Analytics	Vercel Analytics + Mixpanel for user behaviour tracking.

2.2 Admin Dashboard

Layer	Choice & Rationale
Framework	React 19 (Vite). SPA — no SSR needed. Admin is authenticated-only, no SEO requirement.
Language	TypeScript. Shared types package with backend.
Styling	Tailwind CSS v4 + shadcn/ui component library for rapid dashboard UI.
State Management	TanStack Query for all server state. Zustand for UI state (sidebar, modals).
Charts	Recharts for sales charts, bar graphs, and trend lines.
Tables	TanStack Table v8 for sortable, filterable, paginated data tables.
Hosting	Vercel. Separate project from customer site.

2.3 Shared Backend

Layer	Choice & Rationale
Runtime	Node.js 22 LTS
Framework	Express.js with TypeScript
Database	PostgreSQL 16 via Supabase (managed, Lagos-optimised region: us-east-1 closest available)
ORM	Prisma 5. Type-safe queries, migrations, and seeding.
Authentication	JWT (access token 15min + refresh token 7 days). bcrypt for password hashing.
File Storage	Cloudinary for all images. Store only URLs in the database.
Email	Resend (transactional email). Order confirmations, OTPs, delivery updates.
SMS / OTP	Termii. Nigerian phone number verification and delivery SMS.
Push Notifications	Firebase Cloud Messaging (for app — also used as web push on customer site).
Hosting	Railway. Auto-deploy from main branch. Separate staging environment.
Monitoring	Sentry for error tracking. Uptime Robot for uptime monitoring.

3. Database Schema

The PostgreSQL database is managed via Prisma ORM. Below is the complete entity list with key fields and relationships.

3.1 Core Entities

Entity / Table	Key Fields	Notes
users	id, email, phone, name, role, password_hash, loyalty_points, created_at	Customers and admin staff. Role: CUSTOMER | STAFF | ADMIN
addresses	id, user_id, label, street, area, city, is_default	Multiple delivery addresses per user. Magodo Phase 1 pre-filled as default.
products	id, name, slug, description, price, compare_price, sku, stock_qty, category_id, is_mums_pick, origin, images[], is_active	is_mums_pick drives the Mum's Pick gold badge. category_id links to one of 4 categories: Imported, Local, Chilled, Household. A product can be category=Imported AND is_mums_pick=true simultaneously.
categories	id, name, slug, image_url, display_order	Imported, Local, Chilled, Household. The is_mums_pick flag on products is separate from category and drives the Mum's Pick badge independently.
orders	id, user_id, status, subtotal, delivery_fee, total, payment_status, payment_ref, delivery_address_id, rider_id, notes, created_at	Status: PENDING | CONFIRMED | PACKED | OUT_FOR_DELIVERY | DELIVERED | CANCELLED
order_items	id, order_id, product_id, quantity, unit_price, total_price	Snapshot of price at time of order (products can reprice).
shipments	id, name, origin_country, departure_date, arrival_date, status, notes	Tracks US import shipments. Drives pre-order window.
pre_orders	id, user_id, product_id, shipment_id, quantity, status	Customer pre-orders linked to a specific US shipment.
riders	id, name, phone, is_active, current_lat, current_lng	Dispatch riders. Location updated by mobile or WhatsApp bot.
loyalty_transactions	id, user_id, order_id, points, type, description	Points earned and redeemed. Type: EARN | REDEEM | EXPIRE
notifications	id, user_id, title, body, type, is_read, created_at	In-app and push notifications. Type: ORDER | PROMO | SHIPMENT
promo_codes	id, code, discount_type, discount_value, min_order, max_uses, used_count,	Discount type: PERCENTAGE | FIXED. Applied at checkout.

Entity / Table	Key Fields	Notes
	expires_at	

4. API Specification

The REST API is versioned at /api/v1. All responses return JSON. Authentication uses Bearer JWT tokens in the Authorization header. Rate limiting: 100 requests/minute per IP for public endpoints, 500/minute for authenticated endpoints.

4.1 Authentication Endpoints

Endpoint	Access	Description
POST /auth/register	Public	Register new customer. Body: {name, email, phone, password}. Sends OTP via Termii.
POST /auth/login	Public	Login with email+password. Returns {accessToken, refreshToken, user}.
POST /auth/refresh	Public	Exchange refresh token for new access token.
POST /auth/logout	Auth	Invalidate refresh token.
POST /auth/verify-phone	Auth	Submit OTP to verify phone number.
POST /auth/forgot-password	Public	Send password reset link via Resend email.
POST /auth/reset-password	Public	Submit new password with reset token.

4.2 Product Endpoints

Endpoint	Access	Description
GET /products	Public	Paginated product list. Query: ?category=imported|local|chilled|household, ?search, ?mums_pick=true, ?page, ?limit. Category and mums_pick filters are independent and can be combined.
GET /products/:slug	Public	Single product detail by slug.
GET /products/featured	Public	Returns is_mums_pick=true products for homepage hero section.
GET /categories	Public	Returns the 4 active categories: Imported, Local, Chilled, Household — each with product count.
POST /products	Admin	Create new product. Multipart form with images uploaded to Cloudinary.

Endpoint	Access	Description
PATCH /products/:id	Admin	Update product fields including stock_qty and is_active.
DELETE /products/:id	Admin	Soft delete (sets is_active=false). Never hard delete.

4.3 Order Endpoints

Endpoint	Access	Description
POST /orders	Auth	Create order. Body: {items[], delivery_address_id, promo_code?}. Returns order with Paystack payment link.
GET /orders	Auth	Customer's order history. Admin: all orders with filters.
GET /orders/:id	Auth	Order detail with items, address, rider, timeline.
PATCH /orders/:id/status	Staff	Update order status. Triggers push notification to customer.
POST /orders/:id/assign-rider	Staff	Assign dispatch rider to order.
POST /orders/webhook/paystack	Public	Paystack payment webhook. Verify signature, update payment_status.

4.4 Shipment & Pre-Order Endpoints

Endpoint	Access	Description
GET /shipments	Public	Upcoming and recent shipments. Drives pre-order UI.
GET /shipments/:id/products	Public	Products available for pre-order on this shipment.
POST /pre-orders	Auth	Place a pre-order. Links user to product+shipment.
GET /pre-orders/mine	Auth	Customer's active pre-orders.
POST /shipments	Admin	Create new shipment record.
PATCH /shipments/:id	Admin	Update shipment status and arrival date.

4.5 Admin-Only Endpoints

Endpoint	Access	Description
GET /admin/dashboard	Admin	Summary stats: today's revenue, order counts, low-stock alerts, weekly chart data.
GET /admin/customers	Admin	Paginated customer list with order count and total spend.
GET /admin/analytics	Admin	Revenue by date range, top products, category breakdown.
POST /admin/promo-codes	Admin	Create promo code.
GET /admin/inventory/low-stock	Admin	Products where stock_qty is below reorder threshold.
POST /admin/notifications/broadcast	Admin	Send push notification to all or filtered customers.

5. Customer Website — Page Specifications

5.1 Page Inventory

Route	Page Name	Key Content / Features
/	Homepage	Hero with Mum's Picks carousel, category grid, featured local staples, how-it-works strip, delivery zone map.
/shop	Product Listing	Filterable, searchable product grid. Category sidebar. Sort by price, newest, popular.
/shop/[slug]	Product Detail	Image gallery, description, origin badge, add to cart, related products, stock status.
/cart	Cart	Item list with quantity controls, promo code input, order summary, delivery fee calculation.
/checkout	Checkout	Address selector or new address form, delivery time slot picker, Paystack payment.
/orders	Order History	Customer's past and active orders with status timeline.
/orders/[id]	Order Detail	Full order detail with items, rider tracking link, receipt download.
/account	Account	Profile, saved addresses, loyalty points balance, notification preferences.
/mums-picks	Mum's Picks	Dedicated page for US-sourced items. Includes next shipment countdown.
/pre-order	Pre-Order	Browse upcoming shipment products and place pre-orders.
/track/[id]	Order Tracking	Public tracking page (no login needed) via order ID + phone number.
/login	Login / Register	Email/password and Google OAuth. Redirect to /account on success.

5.2 Homepage Layout Detail

The homepage is the primary brand statement. It must load fast (target LCP under 2.5s) and immediately communicate the Mama's Pantry story.

- Navigation bar: Logo left, nav links centre (Shop, Mum's Picks, Imported, Local, Delivery Info), cart icon + account icon right. Sticky on scroll with Forest Deep (#1B4332) background.
- Hero section: Full-width banner with headline 'From our hands to your table', tagline, and CTA button. Show latest US shipment arrival teaser if active ('Mum just landed — new stock in').
- Mum's Picks horizontal scroll: 5-6 product cards with gold Mum's Pick badge. ViewAll link to /mums-picks.
- Category grid: 4 category tiles (Imported, Local, Chilled, Household) with icon and product count. Mum's Picks is a separate featured section driven by is_mums_pick=true, not a category tile.
- Local Staples strip: 4 featured local products. Different visual treatment from imported section.
- Delivery zone section: Simple map or area list showing Magodo Phase 1, Phase 2, Alapere, Ketu, Ojota covered zones.
- How it works: 3-step icon strip (Browse, Order, Delivered to your gate).
- Footer: Logo, links, address (Magodo Phase 1 Gate, Lagos), WhatsApp link, social links, copyright.

5.3 Cart and Checkout Flow

- Cart is client-side (Zustand) and persisted to localStorage. On login, merge guest cart with server cart.
- Checkout step 1: Select delivery address or add new one. Validate Magodo-area addresses for same-day eligibility.
- Checkout step 2: Select delivery time slot (Today 2-6pm, Today 6-9pm, Tomorrow AM, Tomorrow PM). Disable Today slots if order is after cut-off time.
- Checkout step 3: Order summary + promo code field + Paystack payment button.
- On Paystack success: redirect to /orders/[id] with success state. Backend processes webhook independently.
- On payment failure: stay on checkout, show error, allow retry.

6. Admin Dashboard — Page Specifications

6.1 Page Inventory

Route	Page Name	Key Features
/dashboard	Dashboard	KPI cards (today revenue, orders, low stock), recent orders table, weekly sales bar chart, quick actions.
/orders	Orders	Filterable orders table. Filter by status, date, rider. Bulk status update. Click row for detail.
/orders/[id]	Order Detail	Full order view. Status update dropdown, rider assign, customer info, item list, payment status.
/inventory	Inventory	Product table with stock qty, price, active toggle. Low-stock items highlighted red.
/inventory/new	Add Product	Product creation form: name, slug, description, price, images, category (Imported/Local/Chilled/Household), origin, Mum's Pick toggle (is_mums_pick boolean, independent of category).
/inventory/[id]	Edit Product	Same as Add Product, pre-filled. Includes stock adjustment field.
/shipments	Shipments	US import shipment tracker. Create shipments, attach products, update arrival status.
/customers	Customers	Customer table with name, phone, orders count, total spend, loyalty points.
/customers/[id]	Customer Detail	Order history, addresses, loyalty transactions, contact info for one customer.
/analytics	Analytics	Revenue chart (daily/weekly/monthly), top 10 products, category breakdown pie, delivery zone heatmap.
/riders	Riders	Active riders list, current assignments, performance stats (deliveries, on-time rate).
/promos	Promo Codes	Active and expired codes, create new code, usage stats per code.
/settings	Settings	Store info, delivery zones, cut-off times, notification templates, staff

Route	Page Name	Key Features
		accounts.

6.2 Dashboard KPI Cards

The dashboard home must load data in under 1.5 seconds. The /admin/dashboard API endpoint returns all KPI data in a single call to avoid waterfall requests.

Widget	Specification
Today's Revenue	Total confirmed + delivered order value for current day. Formatted as ₦xxx,xxx. Shows % change vs yesterday below.
Active Orders	Count of orders in PENDING, CONFIRMED, PACKED, or OUT_FOR_DELIVERY status. Click to jump to /orders filtered by active.
Low Stock Alerts	Count of products where stock_qty is below the reorder threshold (configurable per product, default 10 units). Red colour when > 0.
New Customers	Customers who registered today. Useful for tracking marketing campaign effectiveness.
Weekly Revenue Chart	Bar chart via Recharts. 7 bars (Mon-Sun). Hover tooltip shows daily total. Saturday bar in Harvest Gold as visual signature.
Recent Orders Table	Last 10 orders with columns: Order ID, Customer, Amount, Status badge, Time. Clicking a row opens the order detail drawer.

6.3 Order Management Workflow

- New order arrives (via webhook): status = PENDING. Dashboard shows pulsing badge on Orders nav item.
- Staff opens order, reviews items, clicks 'Confirm Order': status = CONFIRMED. Customer receives push notification.
- Staff packs items, clicks 'Mark as Packed': status = PACKED. Rider is assigned in same action.
- Rider picks up, staff or rider updates 'Out for Delivery': status = OUT_FOR_DELIVERY. Customer SMS sent via Termii.
- Rider marks delivered (via admin mobile view or WhatsApp bot): status = DELIVERED. Loyalty points credited to customer.
- All status changes are logged to an order_events table with timestamp and actor (staff member name).

7. Component Architecture

7.1 Shared Component Library

Create a shared /components/ui directory. Key components that must be built before pages:

Component	Specification
ProductCard	Image, name, price, origin badge (Mum's Pick or Local), add-to-cart button. Used in grid and horizontal scroll.
OrderStatusBadge	Coloured pill for each order status. PENDING=amber, CONFIRMED=blue, PACKED=teal, OUT_FOR_DELIVERY=orange, DELIVERED=green, CANCELLED=red.
PriceDisplay	Formats number as ₦ with thousands separator. Handles NGN only. Never show decimals for whole naira amounts.
StockBadge	Shows 'In Stock', 'Low Stock' (< 10), or 'Out of Stock'. Colour coded.
DeliveryZonePicker	Dropdown of delivery zones with estimated delivery time and fee. Validates if address is in coverage area.
PaystackButton	Wraps Paystack inline SDK. Accepts amount, email, metadata. Emits onSuccess and onClose events.
MumsPickBadge	Gold pill badge 'Mum's Pick'. Used on product cards and detail pages.
OriginBadge	Green 'Local' or teal 'US Import' pill. Applied to all product cards.
ShipmentCountdown	Timer showing days until next US shipment arrival. Shown on Mum's Picks page and homepage.
LoyaltyPointsDisplay	Shows current points balance with conversion to naira value (e.g. 500 pts = ₦500 off).

7.2 Page Layout Components

- CustomerLayout: Sticky navbar, page content, footer. Used by all customer-facing pages.
- AdminLayout: Left sidebar (collapsible), top bar with store name and logged-in user, main content area. Persistent on all admin pages.
- AuthGuard: HOC that redirects unauthenticated users to /login for customer pages and /admin/login for admin pages.
- AdminGuard: HOC that checks user.role is STAFF or ADMIN before rendering admin pages. Redirects to 403 if not.

8. Authentication & Security

8.1 Customer Auth Flow

- Registration: email + phone + password. Phone OTP sent via Termii. Account not active until phone verified.
- Login: email + password. Returns JWT access token (15 min) and refresh token (7 days, stored in httpOnly cookie).
- Google OAuth: via NextAuth.js. Creates account on first login. Links to existing account if email matches.
- Token refresh: client automatically calls /auth/refresh when access token is within 2 minutes of expiry.
- Logout: deletes refresh token cookie, clears Zustand state, redirects to homepage.

8.2 Admin Auth

- Separate login page at /admin/login. No Google OAuth. Email + password only.
- Roles: ADMIN (full access) and STAFF (cannot access analytics, settings, or customer data).
- Session timeout: 8 hours of inactivity. Admin must re-login. Prevents unauthorised access on shared store devices.
- All admin API routes validate role from JWT payload before processing.

8.3 Security Checklist

- HTTPS enforced on all routes. HTTP redirects to HTTPS.
- Paystack webhook signature verification on every webhook call (HMAC-SHA512).
- Input validation: Zod schemas on all API request bodies. Reject malformed requests with 400.
- SQL injection: impossible via Prisma parameterised queries. Never write raw SQL.
- Rate limiting: express-rate-limit on all public endpoints. Stricter limits on /auth routes (10 req/min).
- CORS: only allow requests from mamaspantry.com.ng and admin.mamaspantry.com.ng origins.
- Environment variables: all secrets in .env files. Never commit to git. Use Railway's environment config in production.

9. Deployment & Infrastructure

9.1 Environments

Environment	Infrastructure	Deploy Trigger
Development	Local machine. .env.local files. Local PostgreSQL via Docker.	npm run dev on all three repos concurrently.
Staging	Vercel preview URLs + Railway staging environment. Auto-deployed on PRs.	Use a separate Supabase staging database. Never use production data.
Production	mamaspantry.com.ng + admin.mamaspantry.com.ng + api.mamaspantry.com.ng.	Deployed on merge to main branch. Manual promotion from staging.

9.2 Monorepo Structure

Use a pnpm monorepo (Turborepo) with the following packages:

```
apps/
  web/          — Next.js customer site
  admin/        — React admin dashboard (Vite)
  api/          — Node.js/Express backend
packages/
  types/        — Shared TypeScript types (Order, Product, User, etc.)
  ui/           — Shared UI components (used by web and admin)
  config/       — Shared ESLint, Prettier, TypeScript configs
```

9.3 CI/CD Pipeline

- GitHub Actions: on every pull request, run type-check, lint, and unit tests across all packages.
- Vercel: auto-deploys web and admin apps on merge to main. Preview deploys on PRs.
- Railway: auto-deploys API on merge to main. Runs prisma migrate deploy before starting server.
- Database migrations: never run manually in production. Always via Railway's deploy hook.

10. Performance & SEO

10.1 Core Web Vitals Targets

Metric	Target & Strategy
LCP (Largest Contentful Paint)	Under 2.5 seconds on 4G. Achieved via Next.js image optimisation, Cloudinary CDN, and server-side rendering of product data.
FID / INP (Interaction to Next Paint)	Under 200ms. Achieved by keeping client-side JS minimal. Use React Server Components wherever possible.
CLS (Cumulative Layout Shift)	Under 0.1. All images must have explicit width and height props. Use next/font to eliminate font swap.
TTFB (Time to First Byte)	Under 500ms. Railway API must respond within 200ms for cached dashboard queries. Use Redis for dashboard KPI caching (15 minute TTL).

10.2 SEO Configuration

- Each product page: unique title tag ('Tillamook Sharp Cheddar — Mama's Pantry Lagos'), meta description, and Open Graph image via Cloudinary.
- Homepage: structured data (LocalBusiness schema) with address (Magodo Phase 1, Lagos), opening hours, phone number, and delivery areas.
- Product pages: Product schema with price, availability, and image.
- Sitemap: auto-generated via next-sitemap. Includes all product and category pages. Exclude /checkout, /cart, /account.
- robots.txt: disallow /admin, /api, /checkout, /cart, /account.

11. Build Phases & Timeline

Phase	Deliverables	Milestone
Phase 1 — Foundation (Weeks 1-3)	Monorepo setup, shared types, DB schema + migrations, auth API, basic product CRUD API	Core infrastructure ready. Can run locally end-to-end.
Phase 2 — Customer MVP (Weeks 4-7)	Homepage, /shop, /shop/[slug], cart, checkout (Paystack), /orders, /account, login/register	Customers can browse, order, and pay. Soft launch ready.
Phase 3 — Admin MVP (Weeks 6-9)	Dashboard KPIs, orders management, inventory management, basic analytics	Staff can process orders and manage stock. Overlaps with Phase 2.
Phase 4 — Mum's Picks & Pre-Order (Weeks 9-11)	Shipment tracker, pre-order flow, Mum's Picks page, shipment countdown widget	Core brand differentiator feature live.
Phase 5 — Growth Features (Weeks 11-14)	Loyalty points, promo codes, push notifications, delivery tracking, rider assignment	Retention and engagement features.
Phase 6 — Polish & Launch (Weeks 14-16)	Performance tuning, SEO, browser testing, load testing, staging sign-off, production deploy	Grand opening.

12. Pre-Launch Technical Checklist

A. Security

- All environment variables confirmed in Railway and Vercel production configs
- Paystack webhook HMAC verification tested with live key
- CORS policy confirmed for production domains only
- Rate limiting tested under load
- All admin routes tested with STAFF vs ADMIN role boundary

B. Payments

- Paystack live key (not test key) configured in production
- Payment success and failure flows tested end-to-end
- Webhook receives and processes payment events correctly
- Order status updates to CONFIRMED on payment success
- Failed payment does not create order record

C. Notifications

- Termii SMS sending confirmed on Nigerian phone numbers
- Resend email confirmed: order confirmation, OTP, password reset
- Push notification opt-in flow working on Chrome and Safari
- FCM delivery confirmed on test devices

D. Performance

- Lighthouse score above 90 on homepage (mobile)
- Product images served from Cloudinary CDN
- Database queries under 100ms (check slow query log)
- API responses under 300ms for all critical endpoints

Appendix: Amendment Log

Amendment 1 — Category Structure Revision

The original specification listed 5 categories including 'Mum's Picks' and 'Imported' as separate browseable categories. This has been revised as follows:

Category Structure — Final

Categories (4 total): Imported, Local, Chilled, Household
is_mums_pick: boolean field on the products table — independent of category
A product can be category=Imported AND is_mums_pick=true simultaneously
GET /products?category=imported returns all imported goods
GET /products?mums_pick=true returns all Mum's Pick items regardless of category
The Mum's Picks page (/mums-picks) and gold badge are driven by is_mums_pick, not a category filter
The category filter UI on Browse shows: All, Imported, Local, Chilled, Household
The Mum's Pick toggle on Browse is a separate boolean filter, not a category

End of Web App Technical Specification

Mama's Pantry — Magodo Phase 1, Lagos — mamaspantry.com.ng